

# YASH DUSANE

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## EDUCATION

Master of Science, Computer Science, California State University Fullerton

Aug 2024 — May 2026

Bachelor of Engineering, Computer Engineering, Savitribai Phule Pune University

Aug 2020 — May 2024

## SKILLS

- **Programming:** Python, Java, JavaScript, C++, SQL, HTML/CSS
- **Backend:** FastAPI, Express.js, Node.js, RESTful APIs, Microservices, WebSockets, NoSQL(PostgreSQL, MongoDB)
- **Frontend & Web:** TypeScript, Next.js, React.js, Material-UI, Tailwind CSS, Redux, Progressive Web Apps (PWA)
- **AI/ML:** TensorFlow, Scikit-learn, LangChain, LangGraph, RAG Systems, Vector Databases, NLP, LLMs
- **Cloud:** AWS (ECS, S3, Lambda), Azure (Data Factory, Databricks, Synapse), Docker, CI/CD (GitHub Actions)
- **Tools & Practices:** Git, Agile/Scrum, Test-Driven Development(TDD), Testing (Jest, Pytest), API Documentation (Swagger), System Design, Data Structures & Algorithms, OAuth(Google, Github), JWT, Session Management.

## EXPERIENCE

Software Developer Intern, PESH Group

Jan 2024 --- Jun 2024

- Resolved slow page load times by 45% and supported 5,000+ concurrent users by developing web application with server-side rendering, optimized database queries with indexing, and Redis caching using Next.js, FastAPI, and PostgreSQL.
- Developed an automation framework integrating third-party APIs and LLM services, automating 80% of support queries with sub-2s response times and reducing ticket volume by 50% using FastAPI and RESTful architecture.
- Boosted organic traffic by 60% and reduced bounce rate by 35% by optimizing application performance through code splitting, lazy loading, and CDN integration using Webpack and Lighthouse metrics.
- Designed and implemented automated testing framework with CI/CD pipeline across dev, staging, and production environments, achieving 95%+ test coverage, reducing deployment time by 60% using GitHub Actions, Jest, Pytest.

## PROJECTS

### AI-Driven ESG Compliance Automation Framework

- Architected compliance platform reducing ESG report preparation from 5-6 weeks to under 2 hours via seven-stage pipeline ingesting files, matching entities, validating data, and generating multi-framework reports.
- Engineered data ingestion handling complex Excel files with OpenPyXL, entity matching achieving 90% precision with ML models, and unit normalization converting 50+ measurement types.
- Developed sector-specific validation engine with configurable rules detecting anomalies, implemented confidence scoring routing low-confidence items to FastAPI review interface, maintaining 95% error detection with <5% false positive rate.
- Implemented RAG system using ChromaDB for semantic search, LLM APIs for narrative generation with citations, W3C PROV provenance tracking, and multi-framework export with reconciliation notes.

### Crush My Course - AI-Powered Academic Course Tracker [\[Demo\]](#)

- Created course planning platform helping students navigate 10,000+ CSUF courses with intelligent recommendations based on professor ratings and pass rates, processing queries in <500ms.
- Designed microservices with automated testing framework, Next.js SSR frontend, FastAPI backend (15+ RESTful endpoints), PostgreSQL with optimized indexes, ChromaDB semantic search, and JWT/RBAC authentication.
- Containerized components using Docker, deployed on AWS ECS with Fargate and Application Load Balancer, integrated Redis caching reducing database load by 40% for 500+ concurrent users.
- Deployed semantic search generating sentence transformer embeddings with hybrid retrieval (vector similarity + keyword filtering) and LLM-powered contextual recommendations.

### ElderEase - AI-Powered Digital Companion for Elderly Wellness [\[Github\]](#)

- Launched digital health platform for medication management and healthcare navigation with medication reminders, appointment scheduling, health tracking, and conversational AI support.
- Built microservices with React.js responsive frontend (WCAG compliant), FastAPI backend (20+ RESTful endpoints), Appwrite authentication, and event-driven real-time synchronization across 5+ services.
- Integrated multi-container application using Docker Compose orchestrating services with health checks for automatic recovery, Docker networks with isolation, and comprehensive Swagger API documentation with monitoring dashboard.

### Tiffin Service Marketplace - Progressive Web Application [\[Demo\]](#)

- Architected two-sided marketplace connecting meal providers with customers via PWA supporting registration, menu management, booking, order tracking, and payment with offline-first capabilities.
- Developed full-stack PWA using React.js with service workers (offline functionality), Material-UI for responsive design, Express.js backend (25+ RESTful endpoints), and MongoDB achieving <200ms API response times.
- Secured architecture with Google OAuth, JWT authorization for stateless sessions, role-based access control, input validation preventing injection attacks, and rate limiting for DDoS protection.